TERTIARY INFOTECH ACADEMY PTE LTD
UEN: 201200696W

LESSON PLAN

For

Fundamentals of Semiconductor Device Physics and Process Integration

TGS Ref No: TGS-2024049339

Conducted by

Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

Version 4.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	29 Sep 2024	First version	Tertiary Infotech Academy
2.0	23 Jul 2025	New format	Tertiary Infotech Academy
3.0	17 Oct 2025	Updated organisation name	Tertiary Infotech Academy
4.0	20 Aug 2026	Rebuilt two-day schedule with current slide map, activities and 4:00-6:00 PM assessment	Tertiary Infotech Academy

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Course Overview

A two-day, 16-hour WSQ course comprising 14 hours of classroom facilitation and two hours of summative assessment. The learning sequence moves from device physics and fabrication architecture to process interactions, performance verification, reliability and follow-up action.

Learning Outcomes

- LO1: Identify semiconductor manufacturing process flows based on device physics.
- LO2: Determine follow-up actions for process integration issues and performance.

Day 1 Schedule - Device Physics and Fabrication

Time	Duration	Topic / Activity	Method	Slides
09:00-09:30	30 min	Digital attendance, introductions, outcomes, TSC and assessment briefing	Facilitation	1-15
09:30-10:30	60 min	Semiconductor materials, energy bands, carriers and doping	Lecture / questions	16-39
10:30-10:40	10 min	Morning break	Break	-
10:40-12:10	90 min	PN junctions, diodes, MOS capacitors, MOSFETs and BJTs	Lecture / worked examples	40-106
12:10-12:50	40 min	Lunch	Break	-
12:50-14:20	90 min	Memory, wafer preparation, oxidation, RTP and lithography	Lecture / peer sharing	107-189
14:20-14:30	10 min	Afternoon break	Break	-
14:30-16:00	90 min	CMP, etch, doping, deposition and metallisation modules	Lecture / case prompts	190-258
16:00-17:00	60 min	CMOS process integration flow and design rules	Lecture / cross-section review	259-299
17:00-18:00	60 min	Activities 1-3: physics map, architecture and CMOS flow	Individual / peer review	441-443

Day 2 Schedule - Integration, Verification and Action

Time	Duration	Topic / Activity	Method	Slides
09:00-09:20	20 min	Digital attendance and Day 1 retrieval practice	Facilitation	2, 300
09:20-10:30	70 min	Isolation, wells, self-aligned modules and wafer sort	Lecture / discussion	300-318
10:30-10:40	10 min	Morning break	Break	-
10:40-12:00	80 min	Contamination, yield, SPC and physical metrology	Lecture / worked evidence	319-364
12:00-12:40	40 min	Lunch	Break	-
12:40-13:50	70 min	Device performance and reliability verification	Lecture / failure analysis	365-386
13:50-14:00	10 min	Afternoon break	Break	-
14:00-14:50	50 min	Packaging, final test and emerging integration	Lecture / examples	387-440
14:50-16:00	70 min	Activities 4-8 and integration review	Case study / peer challenge	444-448
16:00-17:00	60 min	Written Assessment (SAQ) - open book	Assessment	-
17:00-18:00	60 min	Case Study Assessment - open book	Assessment	-

Topic-by-Topic Facilitation Map

LU1: Fundamentals of Device Physics and Device Fabrication - Slides 16-299

- Semiconductor materials and carrier physics - Slides 16-39
- PN junctions and diode behaviour - Slides 40-51
- MOSFET and BJT device operation - Slides 52-106
- Memory devices and integrated circuits - Slides 107-123
- Wafer fabrication process modules - Slides 124-258
- CMOS integration flows - Slides 259-294
- Layout and design rules - Slides 295-299

LU2: Process Integration Issues - Slides 300-440

- Isolation, wells and self-aligned modules - Slides 300-315

- Contamination, yield and process variation - Slides 316-364
- Device performance characterisation - Slides 365-374
- Device reliability and failure mechanisms - Slides 375-386
- Assembly, packaging and final test - Slides 387-419
- Optimisation and emerging technologies - Slides 420-440

Activity Facilitation Map

Activity 1: Device Physics Evidence Map - Slide 441

Purpose: Connect device-physics mechanisms to observable manufacturing controls and device consequences.

- Method: individual analysis followed by peer challenge; planned duration 35 minutes.
- Evidence: Completed device-physics evidence map with at least six mechanism-control-consequence links.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 2: Process Module Architecture - Slide 442

Purpose: Identify the major process loops, their purposes, inputs, outputs and control gates.

- Method: individual analysis followed by peer challenge; planned duration 40 minutes.
- Evidence: Process architecture map with module boundaries, control gates and two feedback loops.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 3: CMOS Manufacturing Flow and Control Gates - Slide 443

Purpose: Outline a coherent CMOS process flow and show where device physics constrains sequencing.

- Method: individual analysis followed by peer challenge; planned duration 50 minutes.
- Evidence: One-page CMOS process flow with control gates and three device-physics sequence rationales.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 4: Lithography-Etch Interaction Case - Slide 444

Purpose: Determine how variation propagates between lithography, etch, clean and contact modules.

- Method: individual analysis followed by peer challenge; planned duration 45 minutes.
- Evidence: Interaction matrix, causal chain and discriminating verification plan.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 5: Doping and Thermal-Budget Interaction - Slide 445

Purpose: Analyse implantation, activation, diffusion and oxide interactions across the thermal budget.

- Method: individual analysis followed by peer challenge; planned duration 45 minutes.
- Evidence: Thermal-budget comparison, evidence map and split-lot experiment design.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 6: Yield and Performance Metric Verification - Slide 446

Purpose: Select and interpret metrics that verify process-integration performance.

- Method: individual analysis followed by peer challenge; planned duration 55 minutes.
- Evidence: Metric dashboard worksheet with calculations, spatial interpretation and verification conclusion.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 7: Reliability Failure Analysis - Slide 447

Purpose: Verify whether TDDB, hot-carrier degradation or electromigration best explains the evidence.

- Method: individual analysis followed by peer challenge; planned duration 45 minutes.
- Evidence: Failure-mechanism comparison and reliability verification statement.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 8: Follow-Up Action and Integration Review - Slide 448

Purpose: Determine proportionate containment, corrective, preventive and verification actions.

- Method: individual analysis followed by peer challenge; planned duration 60 minutes.
- Evidence: Integration review action register with containment, CAPA, owners and effectiveness checks.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Resources Required

- Current trainer slides and learner slides
- Learner Guide
- Individual activity folders and printable PDFs
- Laptop with spreadsheet/PDF viewer
- Projector or display, whiteboard and markers
- LMS access and SSG digital-attendance QR code

Assessment

Assessment runs from 4:00 PM to 6:00 PM on Day 2: one hour for the open-book Written Assessment (K1-K2), followed by one hour for the open-book Case Study (A1-A5). Learners complete assessment digital attendance, submit answers through the LMS and sign the Assessment Summary Record.

Assessment Administration Checklist

- Confirm candidate identity, assessment digital attendance and the approved assessment version before distribution.
- Brief candidates on the one-hour duration, open-book condition, individual work requirement and C / NYC grading outcome for each instrument.
- Issue only candidate papers. Keep answer keys trainer-only and do not place answer-key links in learner-facing systems.
- Collect the Written Assessment before issuing the Case Study, then verify all pages and uploads are attributable to the candidate.
- Record K1-K2 evidence from the Written Assessment and A1-A5 evidence from the Case Study; document reassessment decisions where required.

LMS: https://lms-tms.tertiaryinfotech.com/